

Chapter 3: Congenital Genitourinary Abnormalities — Revision Notes

Williams Obstetrics 26e, pp. 94–145 of the PDF. High-yield numbers are in **bold**. The three tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 9 figures are included from the PDF.

Big picture: The gonads, müllerian ducts and external genitalia come from different primordia, closely linked to the urinary tract and hindgut. Faulty embryogenesis → DSDs and müllerian anomalies that cause infertility, miscarriage, preterm birth and malpresentation; renal anomalies travel with them.

1. Genitourinary Tract Development

Urinary system

- **Weeks 3–5:** intermediate mesoderm → **urogenital ridge** → **genital ridge** (gonads) + **nephrogenic ridge**.
- Each nephrogenic ridge → mesonephros, **mesonephric (wolffian) duct** and **paramesonephric (müllerian) duct**.
- **Weeks 4–5:** each mesonephric duct gives a **ureteric bud** → induces the **metanephros** (final kidney). The bud also forms the metanephric duct (**ureter**).
- Metanephros **ascends to its final position by week 9** (disproportionate caudal growth).
- **Cloaca** (common urinary, genital and alimentary opening) is divided by the **urorectal septum by week 7** → hindgut + **urogenital sinus**.

Urogenital sinus part	Forms
Cephalad (vesicle)	Bladder
Middle (pelvic)	Female urethra
Caudal (phallic)	Distal vagina , Bartholin and paraurethral glands

- Near the end of the 1st trimester the mesonephros degenerates; **without testosterone the mesonephric ducts regress**.
- The ureterovesical junction forms at the **trigone**; faults → obstruction and **vesicoureteral reflux**.

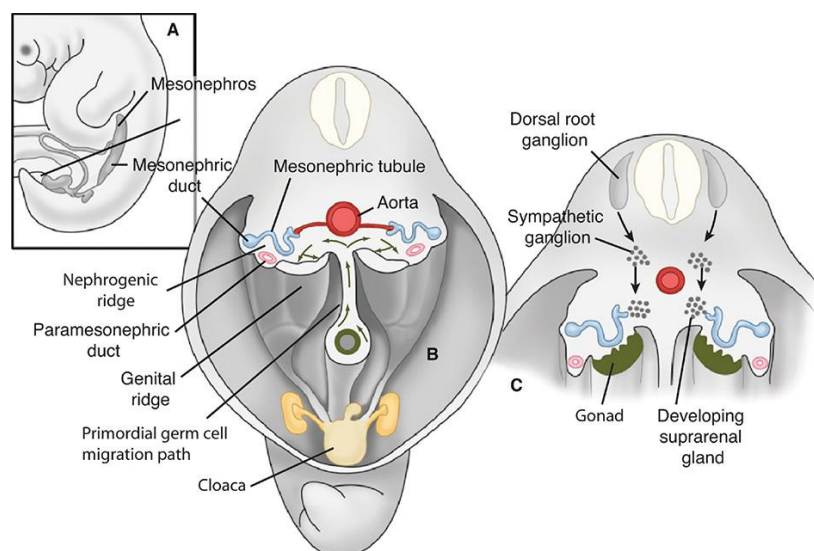


Figure 3-1. Embryo at 4–6 weeks: nephrogenic and genital ridges, müllerian duct, and primordial germ cells migrating from the yolk sac to the genital ridge.

Genital tract

- **Müllerian ducts** → fallopian tubes, uterus, upper vagina. Patterning is guided by **Hox genes**.
- Ducts fuse in the midline → **uterus by ~week 10**. Fusion starts in the **middle** and extends caudally and cephalad.
- The upper wedge-shaped **septum is resorbed**; the final cavity forms **by week 20**.
 - **Failed fusion** → two separate horns (didelphys, bicornuate).
 - **Failed resorption** → persistent **septum**.
- **Vagina**: the fused ducts contact the urogenital sinus → **sinovaginal bulbs** → **vaginal plate** → resorbs to form the lumen; **canalization complete by week 20**. The **hymeneal membrane** separates it from the sinus and degenerates to the hymeneal ring.
- **Nearly half** of females with uterovaginal malformations have **urinary tract defects** (close wolffian–müllerian association).
- With müllerian anomalies, **ovaries are functionally normal** but more often maldescended.

Wolffian remnants in females

Remnant	Site
Gartner duct cyst	Proximal anterolateral vaginal wall (any level). Usually asymptomatic, benign; usually no excision or drainage
Epoöphoron	Blind tubules in the mesovarium
Paroöphoron	Similar tubules adjacent to the uterus

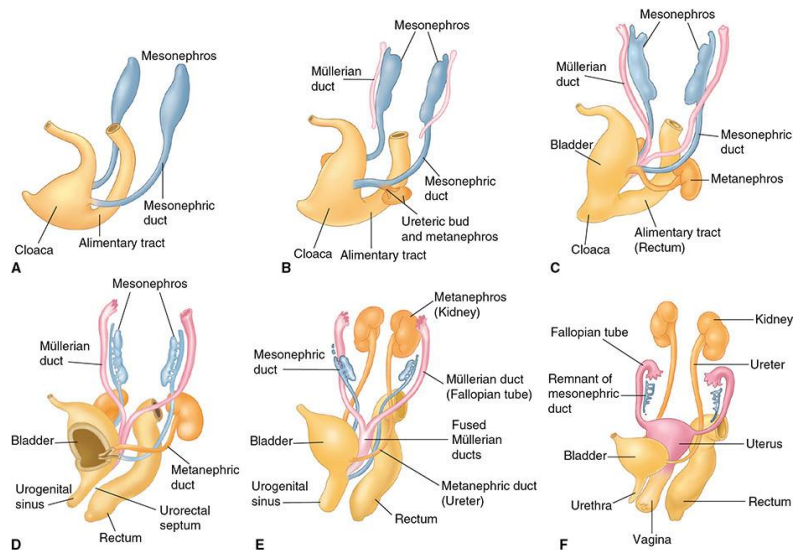


Figure 3-2. Female genitourinary development (A–F): ureteric bud and metanephros, urorectal septum dividing the cloaca, müllerian fusion into the uterus, mesonephric remnants.

Gonads

- **~Week 4**: gonads form from the **genital (gonadal) ridge** = coelomic epithelium on the mesonephros → **primary sex cords**.
- **By week 6**: **primordial germ cells** (future oogonia) migrate **from the yolk sac along the dorsal mesentery** into the genital ridge.
- **Week 7**: sexes distinguishable microscopically. Testis cords → seminiferous tubules and rete testis; mesonephric tubules → efferent ducts; mesonephric duct → epididymis, vas deferens.
- **After week 8**: gonads differ grossly.
- **Female**: primary sex cords → short-lived medullary cords; a second proliferation = **cortical cords** → **primordial follicles by week 16** (oogonium + single layer of follicular cells).
- **By 8 months**: long, narrow, lobulated ovary; tunica albuginea separates coelomic epithelium from the cortex.

External genitalia

- **Week 6:** cloacal folds meet ventrally → **genital tubercle**.
- **Early week 7:** cloacal membrane divides → anal + urogenital membranes; cloacal folds → anal and **urethral folds**; lateral **labioscrotal swellings** appear. **Late week 7:** urogenital membrane ruptures.
- **Visual differentiation not possible until week 12.**
- **Male: DHT** (local **5 α -reduction** of testosterone) → longer anogenital distance, phallus enlarges, labioscrotal folds fuse (scrotum). **Complete by 14 wk.**
- **Female:** no DHT → no fusion. **Differentiated by 11 wk.**

Primordium	Female	Male
Genital tubercle	Clitoris	Glans penis
Urethral folds	Labia minora	Floor of penile urethra
Labioscrotal folds	Labia majora	Scrotum
Urogenital sinus	Vestibule	—

- **First-trimester sex by sonography:** angle of the genital tubercle to a horizontal line along the lumbosacral skin.
 - **>30° = male; <10° = female.**

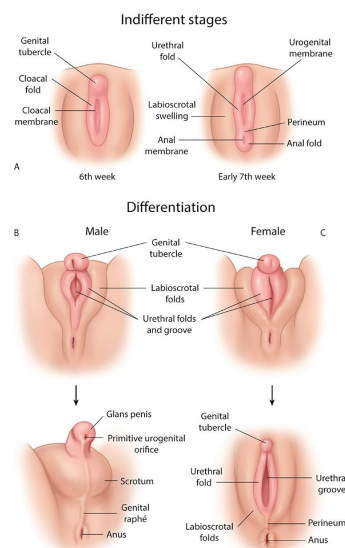


Figure 3-3. External genitalia: indifferent stage (weeks 6–7), then virilization (male) or feminization (female).



Figure 3-4. First-trimester sonogram of a male fetus: phallus angled >30° above the lumbosacral horizontal.

2. Sexual Differentiation

- **Genetic gender** (XX/XY) at fertilization. Embryos look identical for the **first 6 weeks**.
- **Gonadal gender**: Y present → testis, directed by **SRY** (short arm of Y). Also needs autosomal **SOX9, WT1, DAX1, WNT4, NR5A1 (SF1)**.
 - **46,XX male**: Y fragment with SRY translocated to X during paternal meiosis.
 - **46,XY female**: SRY mutation.
- **Phenotypic gender** begins at **8 weeks**. Male differentiation depends on testicular function; **without it, female development ensues** regardless of genetic sex.

Testicular hormones

Cell	Product	Timing	Effect
Sertoli	AMH (müllerian-inhibiting substance)	From 7 wk ; regression complete 9–10 wk	Paracrine → müllerian duct regression (no uterus, tubes, upper vagina). Acts locally : absent testis on one side → ipsilateral horn and tube persist
Leydig	Testosterone (hCG first, then fetal LH)	After Sertoli cells differentiate	Wolffian → vas, epididymis, seminal vesicles. Converted to 5α-DHT → virilizes external genitalia
Leydig	Insulin-like factor 3	—	Testicular descent via the gubernaculum

Table 3-1. Embryonic Urogenital Structures and Their Adult Homologues

Indifferent Structure	Female	Male
Genital ridge	Ovary	Testis
Primordial germ cells	Ova	Spermatozoa
Sex cords	Granulosa cells	Seminiferous tubules, Sertoli cells
Gubernaculum	Uteroovarian and round ligaments	Gubernaculum testis
Mesonephric tubules	Epoöphoron, paroöphoron	Efferent ductules, paradidymis
Mesonephric ducts	Gartner duct	Epididymis, vas deferens, ejaculatory duct
Paramesonephric ducts	Uterus, fallopian tubes, upper vagina	Prostatic utricle, appendix of testis
Urogenital sinus	Bladder, urethra	Bladder, urethra
	Vagina	Prostatic utricle
	Paraurethral glands	Prostate glands
	Greater (Bartholin) and lesser vestibular glands	Bulbourethral glands
Genital tubercle	Clitoris	Glans penis
Urogenital folds	Labia minora	Floor of penile urethra
Labioscrotal swellings	Labia majora	Scrotum

Memory hook: "Skene = prostate, Bartholin = Cowper (bulbourethral)."

3. Disorders of Sex Development (DSD)

- Three categories: **sex chromosome DSD, 46,XY DSD, 46,XX DSD**. Overall **~1 in 5000 births**.

Table 3-2. Disorders of Sex Development (DSD) Classification

Category	Subgroup	Examples
Sex chromosome DSD	—	45,X Turner ^a ; 47,XXY Klinefelter ^a ; 45,X/46,XY mixed gonadal dysgenesis; 46,XX/46,XY ovotesticular DSD
46,XY DSD	Testicular development abnormalities	Complete gonadal dysgenesis; partial gonadal dysgenesis; ovotesticular; testicular regression syndrome
	Androgen production or action defects	Androgen synthesis; androgen receptor; LH/hCG receptor; AMH
	Persistent müllerian duct syndrome	—
	Other structure-affecting syndrome	Smith-Lemli-Opitz
46,XX DSD	Ovarian development abnormalities	Gonadal dysgenesis; testicular; ovotesticular
	Androgen excess	Fetal; maternal; placental

^aAnd syndrome variants. AMH = antimüllerian hormone; hCG = human chorionic gonadotropin; LH = luteinizing hormone.

Definitions

- **Gonadal dysgenesis:** underdeveloped gonads (**dysgenetic testis** or **streak ovary**) → fail → **low sex steroids, high FSH and LH**.
- **Ambiguous genitalia:** hypospadias, undescended testes, micropenis or clitoromegaly, labial fusion, labial mass.
- **Ovotesticular DSD** (formerly true hermaphroditism): ovarian + testicular tissue in one person (any pairing of testis, ovary, streak, dysgenetic testis or ovotestis).
 - Internal ducts follow the **ipsilateral gonad**: inadequate **AMH** → **müllerian structures persist**; inadequate **testosterone** → **ambiguous, undermasculinized genitalia**.
 - Can occur in **all three** DSD categories.

Germ cell cancer (GCC)

- Arises in **dysgenetic gonads with all or part of a Y chromosome**. Candidate gene **TSPY**. Other risks: disturbed gonadal development, delayed germ cell maturation, **gonadoblastoma** (benign; germ cells + immature granulosa cells).
- **Gonadectomy** may be advised by a multispecialty DSD team: early if GCC risk is high or hormones oppose chosen gender identity; later if risk is low and hormones help desired puberty.
- **Undescended testes** also ↑ GCC; **orchiopexy** may be protective.

Sex chromosome DSD

- Sex chromosome aneuploidy **~1 in 2500 births**. Thick **NT or cystic hygroma** ↑ risk; detectable by cell-free DNA.
- **ACMG: cfDNA solely for sex identification without medical indication is not recommended.**

Condition	Frequency	Key features
Turner (45,X)	1 in 2000 female live births (most abort)	Short stature (nearly all); coarctation, bicuspid aortic valve , long QT; renal, skeletal, ENT anomalies; webbed neck (from cystic hygroma); DM, autoimmune thyroiditis, celiac, ↑ LFTs. Two streak ovaries → most common gonadal dysgenesis causing POI . Uterus and vagina normal
Klinefelter (47,XXY)	1 in 650 males	Tall, undervirilized, gynecomastia, small firm testes; reduced fertility. ↑ mediastinal germ cell tumors, osteoporosis, hypothyroidism, DM, breast cancer , CV and neurobehavioral disorders. Thick NT. Pregnancy: preterm birth, cesarean, SGA, neonatal death
47,XXX / 47,YYY	1 / 10,000 females; 2 / 10,000 males	Thick NT, FGR; taller, subtle dysmorphism, autism/ADHD. Hormones and fertility mostly normal (POI or semen abnormalities possible)
Ovotesticular	Rare	46,XX/46,XY (ovary, testis or ovotestis) or mosaic 45,X/46,XY = mixed gonadal dysgenesis (streak + dysgenetic or normal testis). Phenotype varies widely

Turner syndrome in pregnancy

- Conception by **ART** (rarely spontaneous) despite POI.
- Main danger = **aortic dissection** (also AS, hypertension) → MFM counseling + cardiology before conception.
- Use the **aortic size index (ASI)** = aortic diameter / BSA (accounts for short stature).
 - **Avoid conception if ASI >2.5 cm/m²**, or **ASI 2.0–2.5 cm/m² plus risk factors** (bicuspid valve, elongated transverse arch, coarctation, hypertension).
 - **ASRM advises against pregnancy with ASI >2 cm.**
 - Dissection can occur even without risk factors.
- **Preconception work-up:** echocardiography, CT + cardiac MR of heart/aorta, 24-h ambulatory BP, exercise testing, ECG. Serial echos as needed.
- **Risks:** miscarriage, preterm birth, SGA, **preeclampsia, GDM, cholestasis**. Baseline renal, liver, thyroid tests and DM screen.
- **Cesarean rate 40–80%** (complications + cephalopelvic disproportion).

46,XY DSD (formerly male pseudohermaphroditism)

- **Insufficient androgen exposure** in a fetus destined to be male. Testes often present; **uterus usually absent** (AMH works). Usually sterile.

Gonadal dysgenesis

Type	Cause / features
Complete	SRY mutation (Swyer syndrome) or WT1 mutation (Frasier, Denys-Drash). No androgens or AMH → normal prepubertal female phenotype + normal müllerian system
Partial	Intermediate testis development; variable wolffian/müllerian structures and ambiguity
Mixed	Ovotesticular type: one streak gonad + normal or dysgenetic testis
Testicular regression	Testis fails after forming; phenotype depends on timing

- **Gonadectomy often recommended** (GCC risk).

Abnormal androgen production or action

1. **Testosterone biosynthesis defects** → undervirilized male or phenotypic female. **5α-reductase type 2 deficiency** → ↓ DHT, blunted virilization.
2. **hCG/LH receptor defects** → **Leydig cell hypoplasia**, ↓ testosterone.
3. **AMH / AMH receptor defects** → **persistent müllerian duct syndrome (PMDS)**: phenotypic male with a uterus and tubes.
4. **Androgen receptor defects** → **androgen insensitivity syndrome (AIS)**.

Complete AIS (CAIS)

- Phenotypically **normal female**; presents at puberty with **primary amenorrhea**.
- **Breasts develop** (androgen → estrogen conversion); **scant or absent pubic and axillary hair**; **short vagina**; **no uterus or tubes**.
- Testes in the labia, groin or abdomen.
- **Low GCC risk** → testes can be kept **until after puberty** for breast and bone development. Many report mood problems even with hormone replacement after gonadectomy.
- Incomplete AIS → variable virilization; mild forms → severe male-factor infertility.

46,XX DSD

Abnormal ovarian development

Type	Features
46,XX gonadal dysgenesis	Streak gonads, hypogonadism, normal female genitalia and müllerian structures; no other Turner stigmata
46,XX testicular DSD	Usually SRY translocation onto the paternal X . AMH → müllerian regression; androgens → wolffian development and masculinization. No spermatogenesis (Yq genes lacking). Diagnosed at puberty or infertility work-up
46,XX ovotesticular DSD	Unilateral ovotestis + ovary or testis, or bilateral ovotestes. SOX overexpression or deficient ovary-promoting genes

Androgen excess (formerly female pseudohermaphroditism)

- Ovaries, uterus, cervix and upper vagina develop normally → **potentially fertile**. External genitalia virilized according to **dose and timing**: clitoromegaly → posterior labial fusion → phallus with penile urethra.
- Scoring: **Prader 0 (normal female) to 5 (normal male)**; external genitalia score 0–12.

Source	Causes	Notes
Maternal	Luteoma, Sertoli-Leydig cell tumor , virilizing adrenal tumors; testosterone, danazol , androgen derivatives	Tumors rarely affect the fetus: placental aromatase converts C19 steroids (androstenedione, testosterone) to estradiol
Fetal	Congenital adrenal hyperplasia : mostly 21-hydroxylase deficiency (rarely 3β-HSD, 17α-hydroxylase, side-chain cleavage, P450 oxidoreductase, 11β-hydroxylase)	~1 in 18,000 US live births; a frequent cause of virilization
Placental	Aromatase deficiency (fetal CYP19 mutation)	↑ placental androgens, ↓ estrogens → both mother and 46,XX fetus virilized

CAH mechanism and forms

- Enzyme block → ↓ cortisol → ↑ **ACTH** → precursors shunted to androgens.
- **Salt-wasting** (aldosterone blocked, severe): **hyponatremia, hyperkalemia, metabolic acidosis, hypovolemia**: life-threatening.
- **Nonclassic (late-onset, adult-onset)**: unmasked at puberty → hirsutism, acne, anovulation → **mimics PCOS**.
- **Antenatal**: early **maternal dexamethasone** can reduce virilization (Ch 19). **cfDNA fetal sex**: if **Y-chromosome DNA is found, stop dexamethasone** (androgens don't harm a male).

Gender assignment at birth

- A **potential medical emergency** with lasting psychosocial effects.
- Once stable, encourage parents to hold "**your baby**". Use neutral terms: "**phallus**," "**gonads**," "**folds**," "**urogenital sinus**." Explain the genitalia are incompletely formed and that urgent consultation and tests are needed.
- **History**: prior outcomes, medications, consanguinity, antenatal tests and sonography, family history; **maternal hyperandrogenism signs**.

- **Neonatal exam (5 points):**

1. Palpable gonads (labioscrotal or inguinal)
2. Palpable uterus on rectal exam
3. Phallus size
4. Other syndromic features
5. Genital **hyperpigmentation** (↑ MSH with ACTH → suggests CAH)

- **Hyperkalemia, hyponatremia, hypoglycemia → suspect CAH.**

- Tests: genetics, hormones, sonography/MRI (müllerian/wolffian structures, gonads, renal anomalies), ± endoscopy, laparoscopy, gonadal biopsy.

4. Bladder and Perineal Abnormalities

- Without mesodermal ingrowth into the **bilaminar cloacal membrane**, it ruptures early → **cloacal exstrophy, bladder exstrophy or epispadias** (all rare).

Cloacal exstrophy (OEIS complex)

- **O**mphalocele, bladder **E**xstrophy, **I**mpervious anus, **S**pina bifida. **~1 in 300,000** live births.
- Sonography: 1st tri, thin-walled lower abdominal cyst + thick NT; 2nd tri, infraumbilical wall defect, **bladder not seen**, other OEIS defects.
- **Prenatal karyotyping recommended. Delivery route usually dictated by the spina bifida.** Postnatal repair is complex (incontinence, gender assignment issues).

Bladder exstrophy

- Bladder lies open outside the abdomen, drains into amniotic fluid, **never fills**. Abnormal external genitalia, **widened symphysis**; uterus, tubes, ovaries usually normal (occasional fusion defects).
- **Sonographic clues:**
 - **Bladder not seen**
 - Solid mass between the umbilical arteries
 - **Low umbilical cord insertion**
 - Divergent pubic rami
 - **Normal amniotic fluid volume**
 - Males: small penis, anteriorly displaced scrotum
- Differential: bladder exstrophy or agenesis, bilateral ectopic ureters, patent urachus, cloacal exstrophy, simple nonvisualization. MRI helps. Karyotype if genitalia ambiguous.
- **Affected fetus:** routine prenatal course; **cesarean only for obstetrical indications**.
- **Gravida with repaired exstrophy:** ↑ **pyelonephritis, urinary retention, ureteral obstruction, prolapse, miscarriage, preterm birth, breech**. AUA guidelines exist. Some advise **planned early cesarean at a tertiary center** (adhesions); paramedian abdominal and vertical uterine incisions described.
- **Epispadias** alone: patulous urethra, absent or bifid clitoris, unfused labia, flat mons; vertebral anomalies and pubic diastasis common.

Clitoral anomalies

- **Bifid clitoris** (with exstrophy/epispadias); **female phallic urethra** (opens at clitoral tip).
- **Clitoromegaly at birth → excess fetal androgen**. Idiopathic clitoromegaly in extremely premature girls without DSD → observe.

Hymeneal anomalies

- Imperforate, microperforate, cribriform (sieve-like), navicular (boat-shaped), septate: failed canalization of the **inferior vaginal plate (hymeneal membrane)**. The hymen marks the müllerian / urogenital sinus boundary.
- **Hydrometrocolpos (HMC)**: secretions accumulate behind an imperforate hymen.
 - Sonography: **cystic mass behind the bladder**.
 - Most are asymptomatic and **resolve postnatally**. Rare urinary obstruction → **cruciate hymenal incision**.
 - Differential: outlet obstruction (hymen, vagina, cervix), ureterocele, megacystis, ovarian/urachal/mesenteric cyst, anterior meningocele, bowel or bladder duplication, urogenital sinus or cloacal dysgenesis.
 - Rare syndromes: **McKusick-Kaufman, Ellis-van Creveld, Bardet-Biedl**.

5. Müllerian Abnormalities

- Four embryological faults:
 1. **Agensis** of both ducts (focal or complete)
 2. **Unilateral maturation** (other side absent or incomplete)
 3. **Absent or faulty midline fusion**
 4. **Defective canalization**
- **American Fertility Society (1988)** classification is the most widely used.

Table 3-3. Classification of Müllerian Anomalies

Class	Anomaly	Subtypes
I	Segmental müllerian hypoplasia or agenesis	a. Vaginal; b. Cervical; c. Uterine fundal; d. Tubal; e. Combined anomalies
II	Unicornuate uterus	a. Communicating rudimentary horn; b. Noncommunicating horn; c. No endometrial cavity; d. No rudimentary horn
III	Uterine didelphys	—
IV	Bicornuate uterus	a. Complete (division to internal os); b. Partial
V	Septate uterus	a. Complete (septum to internal os); b. Partial
VI	Arcuate	—
VII	Diethylstilbestrol related	—

Data from American Fertility Society, *Fertil Steril* 1988;49(6):944–955.

Memory hook: classes in order "**A U D B S A D**": **A**genesis, **U**nicornuate, **D**idelphys, **B**icornuate, **S**eptate, **A**rcuate, **D**ES.

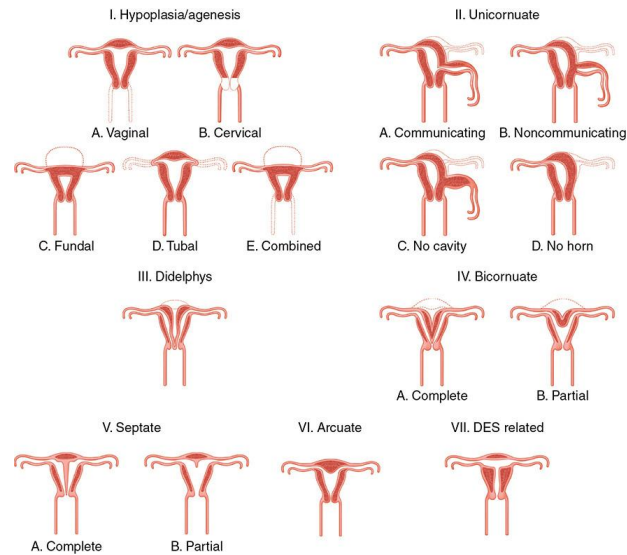


Figure 3-5. AFS classification of müllerian anomalies, classes I–VII with subtypes.

Presentation

- Suspect with a **vaginal septum, blind-ending vagina or duplicated cervix**.
- **Agenesis** → **amenorrhea**. **Outlet obstruction with functioning endometrium** → **pelvic pain** (hematocolpos, hematometra, hematosalpinx) and **endometriosis** (dysmenorrhea, dyspareunia, chronic pain).

Comorbid renal anomalies

- Most often with **unicornuate uterus, didelphys, and ipsilateral obstructing vaginal septum**; less often partial bicornuate or septate.
- **Most common = absent unilateral kidney**.
- Image the urinary tract (sonography, **MRI or IVP**; the last two also show the ureters). Conversely, **renal agenesis** → **image the reproductive tract in early puberty**.

Class I: agenesis

Vaginal anomalies

- **MRKH (Mayer-Rokitansky-Küster-Hauser): upper vaginal agenesis + uterine hypoplasia/agenesis**. Variant **MURCS** (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).
 - A functional vagina can be created, but no childbearing → **IVF with surrogate**; **uterine transplant** is experimental but promising.
- **Complete vaginal agenesis** precludes pregnancy by intercourse unless corrected.

Septum	Labor implications	Management
Longitudinal, complete	Usually no dystocia (fetus descends through one side, which dilates)	Speculum can pass along one side
Longitudinal, incomplete distal	May impede descent	Antepartum resection preferred . In labor: septum thins under the head; with analgesia, clamp, cut and ligate the posterior attachment ; after the placenta, clamp and cut the superior attachment (avoid urethra)
Transverse	Mostly in the lower third ; may be imperforate → obstruction or infertility. In labor a perforate septum can be mistaken for the vault, and its opening for an undilated cervix	After full cervical dilation the head bulges it; stretch the opening; occasionally cruciate incisions (avoid urethra and rectum). Thick septum → cesarean

Cervical anomalies

- Partial/complete agenesis, duplication, longitudinal septum.

- **Complete cervical agenesis = incompatible with pregnancy** → IVF with surrogacy or transmyometrial embryo transfer.
- Uterovaginal anastomosis relieves obstruction but live-birth rates are low and complications (including deaths) occur → some advise **hysterectomy**, reserving reconstruction for selected cervical dysgenesis.

Uterine anomalies: overview

- Prevalence on imaging **0.4–10%**; higher with recurrent miscarriage.
- **Frequency (general population): arcuate > septate > bicornuate > didelphys > unicornuate.**
- As a group ↑ **miscarriage, malpresentation, preterm birth, poor fetal growth.**
- **Vaginal delivery preferred** when feasible: the cervix of the pregnant horn dilates and a proximal septum stretches.

Imaging

Modality	Accuracy / role	In pregnancy
2-D TVS	First line; 90–92%	Yes
3-D TVS	~97% ; coronal views show internal and external contour	Yes
MRI	Up to 100% ; preferred for complex anatomy or planned surgery; shows renal/skeletal anomalies	Yes (without contrast)
Saline infusion sonography	Better internal cavity outline; needs patent cavity	Contraindicated
HSG	Cavity + tubal patency in infertility; poor external contour ; misses noncavitary or noncommunicating horns and obstructed outlets	Contraindicated
Hysteroscopy / laparoscopy + chromotubation	Confirm/treat cavity or tubal disease; screens for endometriosis	Laparoscopy rarely; hysteroscopy contraindicated

Class II: unicornuate uterus

- **~1 in 4000** women. One horn ± rudimentary horn (communicating or not, with or without cavity). Noncommunicating rudiments can lie anywhere along the **paramesonephric migration path** (from the back, sweeping forward along the broad ligament).
- HSG false negatives (noncavitary/noncommunicating horns don't fill) → 3-D TVS or MRI.
- **Renal anomalies in 40%.**

Main horn pregnancy

- ↑ 1st- and 2nd-trimester miscarriage, malpresentation, **FGR**, fetal demise, PROM, preterm delivery (abnormal blood flow, cervical insufficiency, small cavity and muscle mass).
- Median hemiuterus length **5 cm** (normal uterus 7–8 cm). **Internal os–fundus length <4.5 cm** before pregnancy → ↑ preterm labor.
- Surveillance is guided mainly by prior pregnancy outcomes.

Rudimentary horn pregnancy (a cornual ectopic)

- Can occur even in **noncommunicating** cavitary horns via **transperitoneal sperm migration**.
- **Rupture → life-threatening hemorrhage** (converging uterine + ovarian vessels, associated **placenta accreta spectrum**).
 - Most rupture **before 20 wk** (Rolen, 70 cases).
 - Nahum review (588 cases): **half ruptured, 80% before the 3rd trimester; neonatal survival only 6%.**
- **First-trimester sonographic criteria:**
 1. **No continuity** between cervical canal and gestational sac
 2. **Myometrium surrounds** the gestation
 3. PAS-associated **hypervascularity** around the sac

4. **Vascular pedicle** joining the main horn and the sac's myometrium - The main horn shows an empty endometrium continuous with the cervix and an interstitial tube on one side only. 3-D TVS helps.
- **Treatment: excise the horn + pregnancy + ipsilateral tube** (prevent ectopic). Divide uteroovarian ligament, mesosalpinx, round ligament, then the pedicle. Spare the ovary if possible. Route by size and laparoscopic skill.
 - **Nonpregnant:** most asymptomatic; noncommunicating cavity rudiment → obstructive symptoms at puberty ± endometriosis. **Prophylactic excision of any cavity rudiment is recommended.** (Small series after excision: all preterm cesareans.)

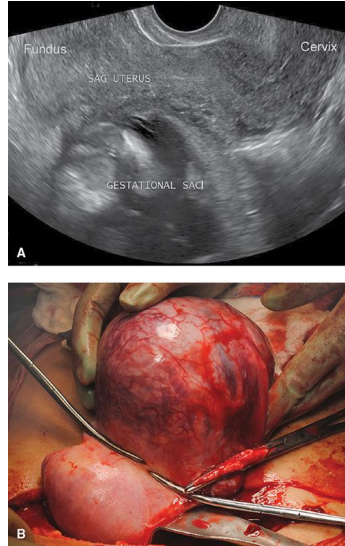


Figure 3-6. Rudimentary horn pregnancy: (A) sac surrounded by myometrium, not continuous with the cervical canal; (B) clamps across the broad vascular pedicle to the main horn.

Class III: uterine didelphys

- **Incomplete fusion** → **two separate hemiuteri, two cervices**, usually two vaginas or a longitudinal vaginal septum.
- **OHVIRA (Herlyn-Werner-Wunderlich):** Obstructed **HemiV**agina + **I**psilateral **R**enal **A**genesis + didelphys. Consider antenatally with **fetal renal agenesis + cystic pelvic mass** (hydrometrocolpos); fetal MRI helps.
- **Diagnosis:** longitudinal septum + two cervices on exam; HSG = two separate canals and fusiform cavities, each with one tube; TVS = divergent horns with a **large fundal cleft**.
- Outcomes like unicornuate but **less frequent:** miscarriage, preterm birth, malpresentation.
- **Metroplasty** (didelphys or bicornuate) only for highly selected unexplained **late recurrent miscarriage** → afterwards **scheduled delivery before labor** (rupture risk).

Class IV: bicornuate uterus

- Fusion anomaly: two hemiuteri; central myometrium extends partly or fully to the cervix.
 - Complete to internal os = **bicornuate unicollis**; to external os = **bicornuate bicollis**. Longitudinal vaginal septum common.
- **Distinguishing bicornuate from septate matters: septate can be treated by hysteroscopic resection.** Use **3-D TVS or MRI** (intercornual angle, fundal contour, interostial line).
- Risks: miscarriage, preterm birth, malpresentation. Metroplasty rarely.

Class V: septate uterus

- **Resorption defect** → complete or partial septum. **Robert uterus:** asymmetric septum → sequestered noncommunicating hemicavity (behaves like a rudimentary horn).
- Found in infertility or **recurrent pregnancy loss** work-ups; 3-D TVS or MRI distinguishes it from bicornuate (criteria still debated).
- Risks: miscarriage, preterm delivery, malpresentation.

- **Hysteroscopic septal resection** may improve birth rates with recurrent loss. Later labor: **small uterine rupture risk**, especially if resection was complicated by **perforation**.

Class VI: arcuate uterus

- Mild deviation from normal. Mostly considered benign; some report ↑ 2nd-trimester loss, preterm labor, malpresentation.

Distinguishing bicornuate, septate and arcuate (Fig 3-7)

Criteria	Bicornuate	Septate	Arcuate
ASRM	Fundal indent >1 cm	Partition depth >1.5 cm; angle <90°	Partition depth <1 cm; angle >90°
Bermejo	Fundal indent <5 mm from the interostial line, or dips below it	Fundal indent >5 mm from the line; partition depth >1.5 cm	Partition depth >1.5 cm (as printed)
ESHRE/ESGE	Fundal indent >50% of uterine wall thickness	Fundal indent flat or <50% of wall thickness; partition depth >50% of above-the-line wall thickness	Partition depth <50% of above-the-line wall thickness

Fig 3-7 prints the Bermejo arcuate criterion as "partition depth >1.5 cm", the same as its septate criterion. It is probably meant to be <1.5 cm.

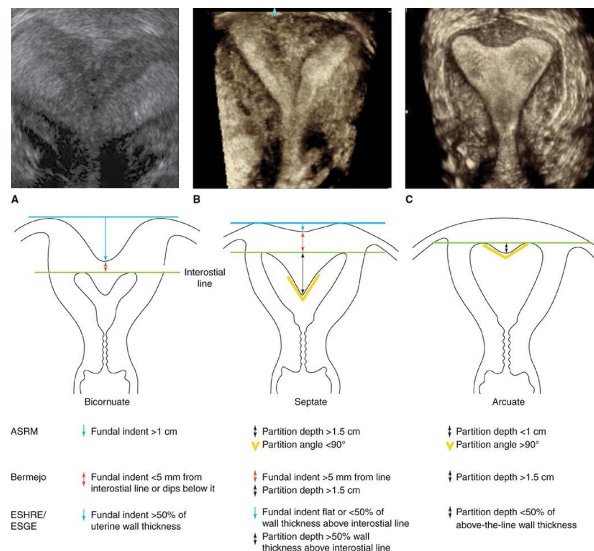


Figure 3-7. Coronal 3-D TVS: bicornuate (external fundal dip to the interostial line), septate (narrow-angled partition, flat fundus) and arcuate (obtuse shallow indent), with ASRM, Bermejo and ESHRE/ESGE criteria.

Cesarean delivery and müllerian anomalies

- **Cesarean rates are ↑**. TOLAC data are few and conflicting. Concerns: small cavity, abnormal myometrial conduction, weaker scar (altered vasculature).
- **ECV is reasonable** to avoid primary cesarean for malpresentation, but a small cavity or midline partition may limit turning.
- **Anomaly first found at cesarean:**
 - Before closure: **explore the cavity** manually (partition length; look for a communication if a rudimentary horn is present).
 - After closure: examine the fundus (**bicornuate vs didelphys** by external contour).
 - **Only one adnexum → suspect unicornuate**; trace the paramesonephric path (back → forward) for a rudiment; remove it or at least **ligate its tube**.
 - **Check both renal fossae** for kidneys; postoperative urinary imaging is reasonable.

Cerclage

- Uterine anomalies are a risk factor for **cervical insufficiency**. Some with repeated post-1st-trimester losses, or partial cervical atresia/hypoplasia, may benefit.
- **Cervical length screening and cerclage use the same criteria as for a normal uterus.**

Class VII: DES-related abnormalities

- **Diethylstilbestrol** (1960s) for threatened abortion, preterm labor, preeclampsia, diabetes: **remarkably ineffective**.
- In utero–exposed women: ↑ **vaginal clear cell adenocarcinoma, CIN, vaginal adenosis**.
- Cervix/vagina: transverse septum, circumferential ridge, **cervical collar**. Uterus: smaller or **T-shaped cavity**. Infertility and adverse outcomes; this cohort is now postreproductive.

Fallopian tube abnormalities

- Tubes form from the **unpaired distal ends** of the müllerian ducts.
- Accessory ostia, complete or segmental agenesis, cystic remnants.
- **Most common = hydatis of Morgagni** (small benign pedunculated cyst at the distal tube). Paratubal cysts may be mesonephric or mesothelial. DES is linked to various tubal anomalies.

6. Uterine Flexion

Anteflexion

- Extreme: the fundus falls forward below the lower symphysis late in pregnancy (lax abdominal wall). Can impair transmission of contractions → **reposition + abdominal binder**.

Retroflexion and incarceration

- A growing retroflexed fundus can become **trapped in the sacral hollow**.
- **Symptoms:** pelvic pressure or pain, **voiding dysfunction or urinary retention**, constipation.
- **Exam:** **cervix anterior, behind the symphysis**; uterus wedged in the deep pelvis. Sonography or MRI confirms.

Management (stepwise)

1. Often **resolves spontaneously in 1–2 wk** with uterine growth; **knee-chest position** several times daily helps.
2. **Indwelling or intermittent self-catheterization** for retention.
3. **Persistent** → **manual repositioning**: empty the bladder, knee-chest position, digital pressure through the **rectum or vagina**; IV sedation or spinal analgesia helps.
4. Then a soft **space-filling pessary for a few weeks** to prevent recurrence.
5. Resistant: colonoscope advancement or colonoscopic insufflation; laparoscopic upward round ligament traction.

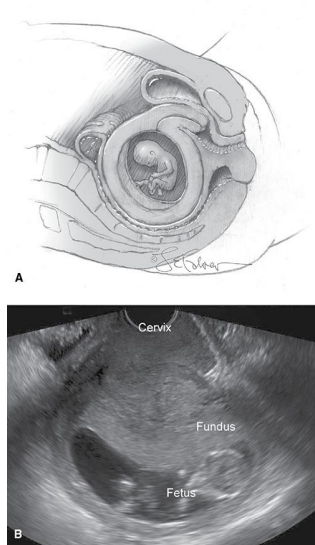


Figure 3-8. Uterine incarceration: retroflexed fundus wedged between promontory and symphysis (A); sonogram with the fundus lying beneath the cervix (B).

Sacculation

- Persistent entrapment → massive **dilation of the lower uterine segment**. The elongated vagina extends above the descended head; the **Foley catheter is palpable above the umbilicus**.
- Sonography and MRI define anatomy.
- **Marked sacculation** → **cesarean** to avoid rupture. Extend the incision above the umbilicus and **deliver the whole uterus before hysterotomy**, restoring anatomy to avoid cutting into the **vagina or bladder**.

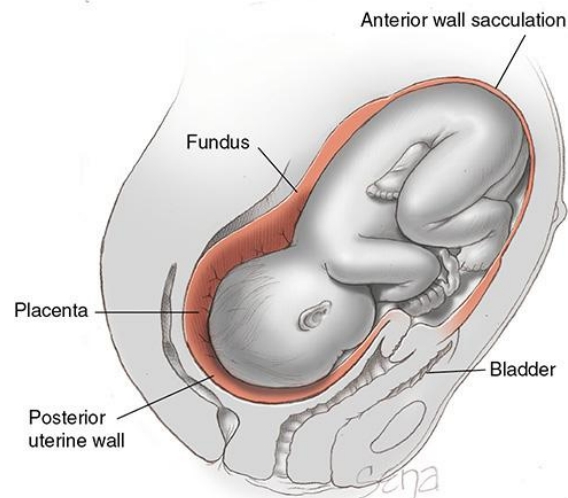


Figure 3-9. Anterior sacculation: attenuated anterior wall and atypical location of the true fundus; beware incising vagina or bladder.

Quick-Recall: Müllerian Anomaly Comparison

Anomaly (class)	Defect	Key feature	Renal link	Obstetric risk / management
MRKH (I)	Agenesis	Absent upper vagina + uterus; normal ovaries	Yes (MURCS)	IVF + surrogate; transplant experimental
Unicornuate (II)	Unilateral maturation	± rudimentary horn; ~1/4000	40%	Main horn: miscarriage, FGR, preterm. Rudimentary horn pregnancy: rupture (50%), excise horn + tube
Didelphys (III)	Failed fusion	2 uteri, 2 cervixes, ± 2 vaginas	Yes (OHVIRA)	Similar to unicornuate but less; metroplasty rarely
Bicornuate (IV)	Failed fusion	Fundal indent; unicollis/bicollis	Sometimes (partial)	Miscarriage, preterm, malpresentation; metroplasty rarely
Septate (V)	Failed resorption	Flat fundus, partition	Sometimes (partial)	Miscarriage, preterm; hysteroscopic resection
Arcuate (VI)	Mild	Obtuse shallow indent	—	Mostly benign
DES (VII)	Drug exposure	T-shaped cavity, cervical collar	—	Clear cell adenocarcinoma, infertility

Quick-Recall: DSD at a Glance

Condition	Karyotype	Gonad	Uterus	External genitalia
Turner	45,X	Streak ovaries	Normal	Female
Swyer (complete gonadal dysgenesis)	46,XY	Streak/dysgenetic	Present	Female
CAIS	46,XY	Testes	Absent	Female (short vagina, breasts, scant hair)
5α-reductase deficiency	46,XY	Testes	Absent	Undervirilized
PMDS	46,XY	Testes	Present	Male
46,XX testicular DSD	46,XX (SRY on X)	Testes	Absent	Male, azoospermic
CAH	46,XX	Ovaries	Present	Virilized; potentially fertile
Mixed gonadal dysgenesis	45,X/46,XY	Streak + testis	Variable	Variable

Key Numbers to Memorize

- Urogenital ridge **3–5 wk**; ureteric bud **4–5 wk**; kidney in place **9 wk**; cloaca divided **7 wk**
- Uterus fuses **~10 wk**; septum resorbed and vagina canalized **by 20 wk**; urinary defects in **~50%** of uterovaginal anomalies
- Germ cells in ridge **by 6 wk**; sexes distinguishable **7 wk**; primordial follicles **16 wk**
- AMH from **7 wk**, müllerian regression **9–10 wk**; phenotypic differentiation from **8 wk**
- External genitalia: female **11 wk**, male **14 wk**, visually distinct **12 wk**; tubercle angle **>30° male, <10° female**
- DSD **1/5000**; sex chromosome aneuploidy **1/2500**; Turner **1/2000** girls; Klinefelter **1/650** boys; XXX **1/10,000**, XYY **2/10,000**; CAH **1/18,000**; OEIS **1/300,000**
- Turner: avoid pregnancy if **ASI >2.5 cm/m²** (or 2.0–2.5 + risk factors); ASRM **>2**; cesarean **40–80%**
- Prader score **0–5**; external genitalia score **0–12**
- Uterine anomalies **0.4–10%**; 2-D TVS **90–92%**, 3-D TVS **97%**, MRI **up to 100%**

- Unicornuate **1/4000**, renal anomaly **40%**; hemiuterus **5 cm** (<4.5 cm → preterm); rudimentary horn: **50% rupture, 80% before 3rd tri, 6% neonatal survival**
- ASRM: bicornuate indent **>1 cm**; septate depth **>1.5 cm**, angle **<90°**; arcuate depth **<1 cm**, angle **>90°**
- Incarceration often resolves in **1–2 wk**